

THE 25 ROBOT COMMANDMENTS

What must a robot establish before it acts?

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Ability to act is not permission to act.

PR-25 sets out 25 candidate assurance requirement families for consequential robot actuation. It describes what must be established, not a particular implementation.

The central rule

NO UNADMITTED CONSEQUENTIAL ACTION.

Within a declared operating domain, fault model, and trust boundary, commanded consequential actuation must stay within a currently valid normal or protective contract. Loss of normal admission must trigger the specified protective response within its justified deadline.

For normal operation, every applicable hard obligation due at the decision must PASS. FAIL, UNKNOWN, or STALE does not confer normal authority. A missing required obligation is not permission. Inapplicability requires an approved profile decision.

Refusal is not the entire safety response

A moving robot may need controlled braking. A manipulator may need to retain a load. A balancing machine may need continued control. Protective behavior therefore needs its own valid contract, operating envelope, and timing argument. It is not a blanket exception or a universal instruction to cut power.

What this release contains

The public paper defines all 25 families, representative evidence and challenge cases, a worked analytical example, and a validation plan. Its separate verification supplement reports an executed finite reference-model check: 3,078 reachable states, 83,106 transitions, and detection of 42 selected faulty variants under the strengthened suite. These results do not verify a physical robot, establish a complete universal registry, or constitute certification.

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25 candidate requirement families

Foundation • PR-01 to PR-07

PR-01 — Hazard-control traceability and closure

Account for declared hazards, allocated controls, evidence, owners, and residual risk.

PR-02 — Assumption and operating-domain validity

Define where assumptions hold, what can be monitored, and what invalidity requires.

PR-03 — Identity and configuration integrity

Match authority to the actual robot, tool, configuration, and relevant payload.

PR-04 — Temporal coherence and freshness

Preserve observation timing, clock constraints, ordering, and validity intervals.

PR-05 — Evidence provenance and integrity

Establish admissible origin, intact content, and evidence dependencies.

PR-06 — Epistemic type and uncertainty integrity

Keep observations, estimates, predictions, assumptions, and their uncertainties distinct.

PR-07 — State-estimation consistency

Require justified state bounds and consistency checks for the control calculation.

Command • PR-08 to PR-13

PR-08 — Authority, purpose, and delegation binding

Limit the action to its authorized target, purpose, mode, and current scope.

PR-09 — Command semantic determinacy

Resolve units, frames, targets, limits, and completion criteria before admission.

PR-10 — Command freshness, replay, and effect integrity

Address expiry, retries, ordering, uncertain outcomes, and repeated physical effects.

PR-11 — Preconditions and transition binding

Bind the action to valid preconditions, an allowed transition, and a defined outcome.

PR-12 — Resource ownership and arbitration

Resolve conflicting control and preserve effective protective precedence.

PR-13 — Bounded interruption, preemption, and required response

Require justified response timing without unsafe interruption or omitted protection.

25 candidate requirement families · continued

Physical safety and assurance · PR-14 to PR-19

PR-14 — Capability and reserve sufficiency

Qualify the resources and margins for normal operation and its protective response.

PR-15 — Safe-envelope and recoverability preservation

Preserve hard constraints and feasible protection throughout execution and takeover.

PR-16 — Protected-entity and exposure constraints

Apply qualified separation, contact, force, access, and exposure limits.

PR-17 — Command-actuation coherence

Compare actual behavior with admitted behavior and justified tracking-error bounds.

PR-18 — Runtime monitor and protective authority integrity

Maintain effective protection within the declared fault and timing assumptions.

PR-19 — Assured recovery and minimum-risk transition

Justify protective feasibility, constraints, objectives, and any claimed completion bound.

Resilience and lifecycle · PR-20 to PR-25

PR-20 — Fault containment and common-cause resilience

Analyze shared dependencies, fault combinations, and propagation.

PR-21 — Cyber and trust-boundary integrity

Protect authority and safety-relevant changes while preserving required protection.

PR-22 — Outcome and audit truthfulness

Distinguish acceptance from verified completion and preserve unknown outcomes.

PR-23 — Controlled adaptation and update integrity

Reassess changes that leave the approved envelope or invalidate dependencies.

PR-24 — Memory, retention, and privacy integrity

Separate safety-state duties from personal-data requirements and resolve conflicts.

PR-25 — Restore, rollback, and version coherence

Restore compatible approved state without reviving revoked authority or repeated effects.

These are summaries. The full paper provides the obligations, evidence examples, challenge cases, phase interpretation, and limitations. A profile must instantiate them for the actual robot and operating domain.

What the framework must demonstrate

Three claims, three kinds of evidence

Admission integrity: does commanded output remain inside the valid contract, under the stated enforcement assumptions?

Modeled physical preservation: do the model, controller, and takeover argument preserve the specified physical constraints?

Application assurance: do the actual robot, environment, faults, measurements, and implementation justify those assumptions?

A result at one level does not establish the next. A signature is not proof of measurement truth. A valid command is not proof of a physical outcome. Denying new commands is not proof that a moving robot can recover.

Evaluate the conditions, not just the headline

Applicability must be explicit. Empty or unresolved required sets cannot silently pass. A future outcome is checked when it can be observed, not assumed at admission. Permissions need a bounded horizon and justified invalidation behavior. Timing must include sensing, computation, communication, arbitration, actuator response, and behavior between checks.

Current evidence status

This public release provides requirements, a validation plan, and executed finite reference-model results. The Verification Note documents the source, reproduction, test-oracle repair, and model limits. No robot or application-validation result is reported. The evidence does not establish that exactly 25 families are complete, irreducible, or sufficient for every application. No certification or standards-compliance claim is made.

Make criticism actionable

Identify the operating profile, assumptions, family or missing obligation, initial conditions, sequence of events, and the output or physical outcome that contradicts the proposed requirement. Distinguish an inadequate requirement from a violated assumption, an implementation defect, or missing evidence.

Counterexamples should change the registry—not be forced into it.

Read the full public paper

PR-25: Candidate Assurance Requirement Families for Robot Actuation, v0.4. The accompanying technical paper contains all 25 full entries, a worked example, open proof obligations, the validation plan, and 16 source references. Its established foundations include runtime assurance, shielding, formal control, hazard analysis, and evidence-based assurance; see references [1–16] in that paper.

STARGA • Public requirements • Application-specific validation remains open.